



Operational planning of the production and the processing of chickens in multi-products slaughter unit

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Abstract

This research studies the production planning of a charcuterie unit that slaughters animal products from poultry, cattle, and sheep farming and processes them into multiple products. The current organization of this unit was carried out in an anarchic manner without prior study or optimization, resulting in significant malfunctions and a high rate of unsatisfied product demands. To reduce this rate, an original contribution was developed based on modeling under realistic constraints, considering the fluctuations in demand over a 1-year planning horizon. An integrated approach was taken, which involved considering several considerations, such as additive production capabilities from overtime or hiring a seasonal team, while respecting the ergonomic aspects of workers in both situations. The obtained results are significantly better than those of the previous organization. This decision-making tool presents a generic model that can be applied to other case studies and has demonstrated its effectiveness in choosing the work mode and determining the optimal quantities of fresh, deep-frozen, and by-product products. The proposed approach has also proven effective in controlling energy consumption at the storage tunnels and providing careful control at the beginning and end of each period. In the end, the developed model was validated with the Cplex solver in a multi-product, multi-period environment, considering several obstacles that translated into constraints. From the modeling point of view, this study can be a valuable resource for the development and extension of other similar cases.

Keywords Production planning · Operational planning · Multi-product · Poultry supply chain

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1 Introduction

The food sector is one of the industrial sectors that have boomed in recent years, especially the poultry sector, which is constantly expanding due to several parameters [1]. On the one side, this expansion is due to population growth, the diversity of products exposed to consumers, as well as technological and biological evolutions that introduce carefully developed techniques and processes for the production, preservation and packaging of food properties. On the other side, the selling price of white meat meets the purchasing power of individuals in the world, particularly in the Third World, where the other meats' prices (red meat and fish) are relatively high and exceed the available income. Unfortunately, this industry fails to satisfy the local market due to the poor production organization within these companies and the misuse of material and human resources, despite the efforts of all actors (companies, farmers, breeders and slaughterhouses) to meet the conflicting market demands [2, 3].

The production and processing of white meat (broilers) is a sequence of operations starting with the reception of the chickens from the breeding houses to the processing units. Each unit has a processing chain starting with the operation of slaughtering, cleaning, cutting the chickens into pieces (formation of by-products: wings, thighs, chicken breasts, etc.), packaging and finally distribution of white meat to consumers. In fact, white meat is a highly perishable product and sensitive to microbes (*Salmonella*) [4], which makes its production difficult, requires a skilled workforce, specialized material resources, and rigorous orderly and dynamic management to avoid the meat contamination throughout the production process.

The management of the poultry chain varies differently from one country to another and from one area to another. This is due to the variability of the considered parameters such as the climate, the available resources (drinking water and food), the adopted processing technology (manual, semi-automatic and automatic), the packaging and preservation at the storage areas and during transport. The non-control of these parameters leads to an imbalance throughout the chain accompanied by an enormous loss of the poultry, causing high deficits and instability of the entire activity in terms of produced quantities and selling prices, and even worse, social unrest resulting from the dismissal of workers when they stop working. This situation has been widely observed, especially for poultry production in Algeria over the last several decades. Despite the availability of local poultry farming networks, different technologies, and workforce, the Algerian market, the poultry sector environment, and the social sector suffer from these repeated disturbances. In fact, the produced quantities of meats vary with price trends [2].

The challenge of this study is to propose a multi-period planning schedule for the chicken transformation into multi-products, applied to a network composed of a group of breeding farms, slaughter units, and storage areas. In the next section, an overview of the fieldwork is presented to position the targeted problem. In particular, this research targets a meat processing unit that ensures the production of several types of products such as fresh chicken, deep-frozen chicken, and by-products, knowing that their production is done anarchically way without a prior plan, which

makes this unit unable to efficiently meet daily demands especially for fresh chicken since this studied semi-automatic meat processing unit chooses to focus on the production of deep-frozen chicken with a single team in regular mode.

Through the description of the problem, the problem addressed is a decision-making-organizational problem in view of the availability of resources, although according to the literature investigated in relation to our point of view no work has dealt in a technical way the modeling of an operational research vision adapted to this type of production system.

In this context, the questions that arise are:

- How to plan, organize and synchronize between different resources to ensure market stability?
- What is the best approach to responding to dynamic and seasonally demands?
- How can disturbances be mitigated to maintain an optimal selling price throughout the production horizon?

To address this problem and develop this industry while ensuring its stability, well-structured coordination and synchronization between all the actors of the poultry supply chain is necessary to serve consumers with healthy meat of good quality.

This research aims to study and analyse the poultry chain behavior, in particular, the supply-processing-distribution relationship applied to a poultry supply chain in an integrated approach. The contribution of this work aims to:

- The development of a generic model purely oriented towards the aggregate planning of meat processing units with the adaptation of real constraints related to the system operation in a mathematical formulation.
- Give more flexibility, facilitation and control to decision-makers to effectively investigate the functional and economic balance sheet of this production unit.
- Make maximum use of production capacity compared to the classic case of other work teams by adding overtime or recruiting other seasonal teams.
- The proposed approach raised the problem of satisfying the demand for the fresh product by reintegrating the optimal production of that product.
- An integrated optimization of the different costs to offer the market the best product in terms of quality-price ratio since this approach also aims to improve the functioning of this system hence the quality of service and product.
- The proposal for the scientific community of a basic technical support that can be improved according to the environment in which the production system is modified compared to the one presented in the article.

The rest of the article is structured as follows: First, a literature review with the contribution of this study are presented in Sect. 2. In Sect. 3, the operation of the studied meat processing unit, the studied problem, and the resolution approach are described. Followed by a detailed mathematical formulation of the case studied in Sect. 4. Section 5 presents the numerical parameters used for the proposed modelling. Then the obtained results from the modelling implementation under the

Cplex solver are presented and analyzed in Sect. 6. Then in Sect. 7 a comparison between the previous unit's management before and after planning, following by a synthesis is Sect. 8. Section 9 present the managerial perspectives. Finally, in Sect. 10 concluding remarks and future research perspectives are provided.

2 Literature review

The problem addressed in this work deals specifically with the implementation and processing of chicken into consumable products, either a whole chicken or chicken divided into different by-products (wings, legs, drumsticks, etc.). This problem calls upon several notions. First of all, it is required to master the functioning of the poultry supply chain and the parameters influencing it [1, 2, 5–9]. Then, an understanding of meat preparation and processing is required for both the poultry and livestock sectors since meat is a highly perishable product, sensitive to microbes and contamination, which makes its processing difficult and requires specific procedures and a skilled workforce.

Recall that, this work targets tactical-operational planning applied to the operation for the processing of poultry products into multi-products in a multi-period framework. Accompanied by the allocation and reorganization of production units. To position this contribution in the literature, a presentation of research work in the field is given below. It should be noted that the research carried out in this field is divided into several parts.

The first part concerns the planning of multi-product production in a multi-period framework. Among the existing works, there are [10–14] who have established planning studies consisting of determining the optimal quantities to be achieved over all periods while minimizing costs. Other researchers have worked on the development and resolution of mathematical models using exact approaches or approximate methods. Also there are [15] presented a book on the optimization and modeling of supply chain problems using different methods of operational research. [16] established a supply chain management by adapting the genetic algorithm to ensure demand satisfaction and cost minimization. [17] developed a multi-criteria mathematical model for the selection of suppliers.

And the second part concerns the organic aspect of the meat production and processing units. Through the literature of works in the field, some researchers have worked on contamination and occupational hygiene, such as [18] who studied the effect of salmonella bacteria in chicken and red meat. In addition, [4] investigated the infection of Salmonella by broiler farmers and slaughter agents. Regarding the white meat quality, some works have been published such as [19–22]. In the other hand, some works have focused on market studies as well as production and consumption statistics that are related to white meat such as in [23–28].

Concerning the meat cutting aspect in general, most of the work has been attributed to red meat (cut of mutton, beef, etc.) and its transformation into consumer products such as minced meat, sausages, etc., or even products in the form of meals and preserves. For red meat, a study by [29] addressed the production and transport

of the various red meat products from the slaughter units to the points of sale. Other works have worked on the machine aspect (design of automatic and semi-automatic machines, what is their impact and what is their evolution, such as [30, 31]).

Regarding white meat and the processes related to its transformation, it is important to mention [32] who wrote a whole book covering all the techniques and stages used for the transformation of poultry into a multi-product. Also, [33] proposed a planning that takes into account several parameters related to the produced quantity and the desired weight. The study made by [34] was carried out based on cutting chickens into specific pieces according to customer requirements and the distribution of white meat to satisfy a fuzzy demand. This study aims to find the best task-machine assignment that maximizes profit within a single-week planning horizon.

Another important notion in this work is the operation reorganization of the production units, which adopt a regular working day of 8 h, six days a week with one day off. However, this may not be sufficient in the face of dynamic, daily, and seasonally changing demands. To meet this challenge and positively deal with the various generated losses, new working methods are proposed.

In fact, the proposed method firstly suggests adding overtime then using a second team while maintaining the regular working method, to satisfy all the received demands. The distribution of tasks and the decision-making choice on the work mode have taken the attention of several researchers. Indeed, an equitable distribution must be found between the use of material and human resources while respecting the tolerated capacities of the equipment (machines, tools and materials) as well as the ergonomic aspect linked to human beings.

Among the researchers interested in the work organization there are some researchers like [35] who carried out a comparative empirical study between two work organization modes; the first mode deals with shift work, and the second mode based on the addition of overtime. [36] established a statistical study to guide the decision-makers in determining the effect of shift work on labor productivity and related costs. [37] provided a comparative study between work management associated with high health risks and work management that respects the ergonomic aspect of workers. [38] analyzed the different factors related to the workers' selection for shift work.

2.1 Contribution

According to above-mentioned research, it can be concluded that the existing works in the context of the tactical-operational planning focus particularly on proposing schedules of production and transformation of the agro-food products in consumables to meet the demands given by nature fruit and vegetables where these products have a purpose intended for consumption as fresh product to satisfy daily sale or conserve them under high degrees of packaging and preservation ([39, 40]), knowing that vegetables and fruits represent more robust characteristics for conservation and they are less sensitive to deterioration than meat.

Other researchers have developed their research by introducing additive modes of operation either in regular plus overtime or in irregular by adopting a second team within the production units. In this context of fruits and vegetables, it is easier to integrate additive hours and reserve them for the realization of this type of products that supports better overtime, which is not necessarily favourable for meat.

Concerning meat in particular, for the poultry chain the majority of works [32–34] have dealt with the problems of: transport, quantities to be processed, a distribution between tasks and machines, storage operations, and cutting operation just in a regular work time, having no recourse to the freezing function exploitable by overtime or additive teams. In addition, they studied several meat processing units similar to the one studied. However, to our knowledge, no scientific work based on operational research applied to planning has dealt with the operation in terms of slaughtering, cutting, by-product processing, deep-freezing, packaging and storage in a multi-period framework. Except for the work of [34] which has contributed to address the cutting problem in a single-period horizon oriented more towards task scheduling. The following table compares the work established in this article with the research cited above to clarify the research gap (Table 1).

The objective of this study is to propose a generic planning model for meat processing unit to define the optimal quantity to produce for each period that meet the given demands, while minimizing stock-outs and overstocking during every period. Specifically, this study proposes different working modes and their most adequate and profitable allocation over the different periods of the planning horizon containing the different constraints. The proposed approach and the model developed in this study, can be adapted to other meat processing units, flexibilities in the production capacity in regular and irregular are suggested to improve the satisfaction rates and reduce the stocking rates, which makes the tactical-operational planning model more generic.

For this purpose, gains are observable at the economic level and the level of the decision-making process resulting from the organization of the unit's functioning. The development of a new work strategy is adapted to the multi-period product, multi-period case by focusing on:

- Realize an integrated approach of tactical-operational planning to better optimize and visualize the different parameters and KPIs of this system to manage the quantities produced (fresh and deep-frozen).
- Propose an approach to reorganize production, making better use of the deep-freezing functions to efficiently satisfy the market.
- Reduce the over stock and rebalance the quantities in stock, by transforming the surplus products into other consumable products

3 Problematic

3.1 Description of the slaughter unit's operation

This research is interested in studying an operational level of a broiler production system composed of a set of farms that ensures the supply of live chickens. They are sent to slaughtering units where they must undergo several transformations. In this context, the chicken will be decomposed into multiple products before being sent to storage areas then finally delivered to final consumers. The studied poultry chain is composed of several elements represented by the following figure (Fig. 1):

The first element concerns the breeding network of various poultry farms that supply the slaughterhouses daily. A previous study was carried out in this chain and which proposed to establish a breeding schedule for all the farms synchronously. This can be done by specifying the start and end dates without overlapping, and assigning the same solicitation priority on the different planning horizons in order to spread to the requests from slaughterhouses [2, 3]. This study has determined the decisional choice related to the poultry sector planning at the tactical level. More precisely, it decides to established a breeding schedule according to the demand quantities of the slaughterhouse.

Based on the obtained results, and to ensure the best progress of the various activities of this sector, this study took place. it aims to propose better planning at the operational level for processing the received chicken from farms by transforming them into ready-to-consume multi-products. Obviously, the production section requires careful and synchronized planning between different operations (slaughtering, de-feathering, cleaning, sorting, packaging, etc.) to minimize the production time in the face of the perishability constraint while maintaining the nutritional characteristics of the white meat.

It is clear that the slightest management error can generate enormous losses since this sector does not tolerate non-productive waiting times, and the least waste in terms of time, workforce and conservation is very valorising for the whole chain. This work focuses on the second element of the chain which deals with the planning and the management of slaughtering units and chicken processing into several by-products, as shown in the following figure (Fig. 2).

The different phases of the production process are detailed in the following figure (Fig. 3).

After receiving the chickens from the farms, the transformation and production process begins: slaughtering, de-feathering, evisceration and cleaning. In this first phase, the chicken is husked and separated into three main groups:

The first group concerns the products obtained from the slaughter (first product: livers, gizzards, and necks). Where the amounts of each by-product type in this group are cleaned, packaged and shipped to the specified customers.

The emptied chicken goes through drying, then sorting, which is the most delicate action since it is at this point that the chicken undergoes a specific processing line after a selective visual test. This test separates the chickens according to general

shape into two groups (2nd and 3rd group), where each group refers to a workstation composed of several production lines.

The second group the chickens of this group are sent to the first work station, which is one unit with two production lines.

This unit receives the chickens that have not suffered any deterioration. At this time, a second sorting is carried out by separating the two chicken qualities: high quality (high weight) and medium quality (medium weight). Then, each quality is sent to a dedicated work line where numerous operations are performed. The first line received the first quality chickens (superior quality), where they are subjected to bagging and packaging operations to obtain “the second product” (fresh chickens) before being sent to the sales outlets to meet daily consumption. The second line prepares the second-quality chicken to produce “the third product” (deep-frozen chickens) after bagging the medium chickens and deep-freezing them in a deep-freezing tunnel. This operation aims to preserve the chickens’ nutritional characteristics. Finally, packaging operation takes place in order to ensure further sales.

The third group includes all chickens with fractures. Indeed, these fractures are due to the chickens’ transport from the breeding farm to the processing unit as well as the crowding of the chickens inside the cage which causes various breaks in the wings, legs and thighs. To this fact, chickens with fractures cannot be sold as whole chickens. This category of chickens will be processed and transported to the second station, which processes these chickens into multi-products ready for sale and marketing. This station is composed of three lines, where each line is dedicated to the production of one product type.

The first line is specialized in cutting chickens into pieces. This process results in “product 4” which includes all pieces divided into five classes: the wing class, the legs class, the breast class, the cutlet class and the carcass class. Knowing that, one chicken gives two wings, two legs, one breast, and one carcass. In this phase, these pieces are sorted according to their type then placed in 1 kg trays. The resulting trays are packaged and labelled.

The unit also has a charcuterie unit for the production of elaborated products “product 5” and “product 6” which are, respectively, chicken jambon with its different variants (chicken jambon, cheese chicken jambon, smoked chicken jambon, etc.), and sausages with their different variants (fresh, vacuum-packed). The production of these by-products is carried out following specific techniques for each by-product.

This work studies the operation of a slaughterhouse composed of several work units where each unit has one or more production lines for the transformation and production of several poultry by-products and their derivatives from live chickens. As mentioned earlier, the vivacity character represents a strong constraint for all the different livestock sectors, in addition to the second constraint of perishability related to the characteristics of fresh products, particularly for the poultry chain. Knowing that, chicken is a very sensitive and highly perishable product. This makes its production and transformation difficult to manage and requires careful and precise management and planning to ensure that the consumable properties of the

Table 1 Literature review

Authors	Planning		Planning horizon		Poultry chain links			Products						
	Strategical	Tactical	Operational	One period	Multi periods	Hatchery	Breeding farms	Slaughterhouse			Distributors	One product	Multi products	
								Slaugh- terhouse location	Production manage- ment	Multi shift working				Meat pro- cessing
Boudahri et al. [7]	X			X				X			X			
Boudahri et al. [8]		X		X					X			X		
Gordon 2013	X	X		X			X		X			X		
You and Hsieb [5]	X	X			X		X					X		
Naprom et al. [33]			X	X							X		X	
Leksakul and Apiromchaiyakul [34]			X	X							X		X	
Prasertatassang et al. [9]		X			X		X				X		X	
Brevik et al. [6]	X	X		X	X		X					X		
Satir and Yaldirim [1]	X			X	X		X					X		
Tahraoui et al. [2]	X			X	X		X					X		
Present work		X	X		X				X				X	

obtained products are maintained in the best possible conditions in order to meet the dynamic market demands and fluctuations (Figs. 4, 5, and 6).

3.2 The problematic

The problem of this poultry chain is the poor organization and desynchronization between production and market demands. Knowing that, these demands are very variable during the year and differ from one period to another. This is due, on the one hand, to the random nature of demands and, on the other hand, to the existence of several types of events: local festivities, conferences, weddings, etc. which leads to a large consumption of chickens compared to regular periods when chicken consumption is more or less stable around a certain average (Fig. 7). This variation significantly influences the operation of the slaughter units since these demands are daily. In addition, the slaughterhouses have an estimated daily production capacity within 8 working hours. When demand is very high, these units are unable to meet all of the demands in an 8-h shift. This problem affects the notion of loyalty and leads to the risk of losing customers.

An additional constraint to this problem is the management of the cold storage room. Since each deep-frozen chicken has to be stored for 48 h before being sold and consumed to satisfy the daily demand for this product, the system requires deferred production. More precisely, the product delivered and sold on day "d" has to be ready and available at least 48 h before, i.e. the duration has to be longer than "d-2" according to the deep freezing standards. This principle forces the company to have previous safety stocks to satisfy current demand and requires the replenishment of the stock regularly to satisfy future demands.

In this context, and to overcome the problem of unsatisfied demands and their consequences, this study proposes a flexible work schedule between the different work modes presented (regular work, overtime work and second shift work).

3.3 Resolution approach

The chicken production and transformation network operates on a regular 8-h work day, six days a week, throughout the year.

This study has introduced other work modes through overtime or the recruitment of a second team. Note that, to ensure the ergonomic conditions of the worker, overtime is limited and cannot exceed a certain value (for example, from 1 to 4 h maximum per day). Thus, it proposes to add a second shift (2×8 H per day). This adds further constraints to the system since the requirement for this team is that it will be recruited for a minimum of one month's work, with repetitive contracts as needed. This gives three different work configurations. Firstly, work with the regular mode where employers will work daily for 8 h. Secondly, regular work mode with overtime. Finally, regular work mode with a second shift if necessary. Within this framework, the working hours for each work mode are presented in the following table (Table 2).

More precisely, the first configuration consists in a regular shift where 8 h of work are tolerated each day. The second configuration (regular shift+overtime) allows

workers to add overtime beyond the 8 h that are tolerated each day. This additional time is limited by a maximum of three hours to ensure the respect for the workers' ergonomic aspects.

In the case where demands are very high, the third configuration is adopted. It employs a second shift: the first shift works for 8 h (first working period), and the second shift starts when the first shift finishes (to work the next 8 h). The only constraint for this configuration is that the second shift cannot be hired for a day or two but rather on a fixed-term contract.

The objective of this study is to determine the quantity to be produced to satisfy demands and the corresponding work mode for the production of this quantity.

Thus, the considered problem is composed of two different decisions. On the one hand, it is required to determine the quantity to be produced to satisfy the demand, and on the other hand, the adequate work mode must be defined. To solve this problem, a mathematical model has been proposed followed by a resolution by the Cplex solver.

4 Mathematical modelling

During a visit to the slaughterhouse and the poultry chamber of Tlemcen-Algeria, several reflections were communicated by the various met officials and managers. Several assumptions have been made to respond to these reflections, and facilitate the modelling of the target problem. According to the operation mode of the studied link, which is the slaughterhouse, these hypotheses are defined after a collection of data from visits to the poultry company as well as field trips, which permitted the gathering of all the information related to the studied problem. The main points to consider are grouped under the following assumptions:

- The demands of each period are grouped and transferred to the slaughterhouse weekly.
- The study is done in one slaughterhouse.
- The regular work team accepts an additional capacity of 1 h to a maximum of 4 h (1500 and 8000 chickens).
- The second team is recruited only in periods of very high demand. For periods of low demand, the first shift is maintained.



Fig. 1 The poultry supply chain

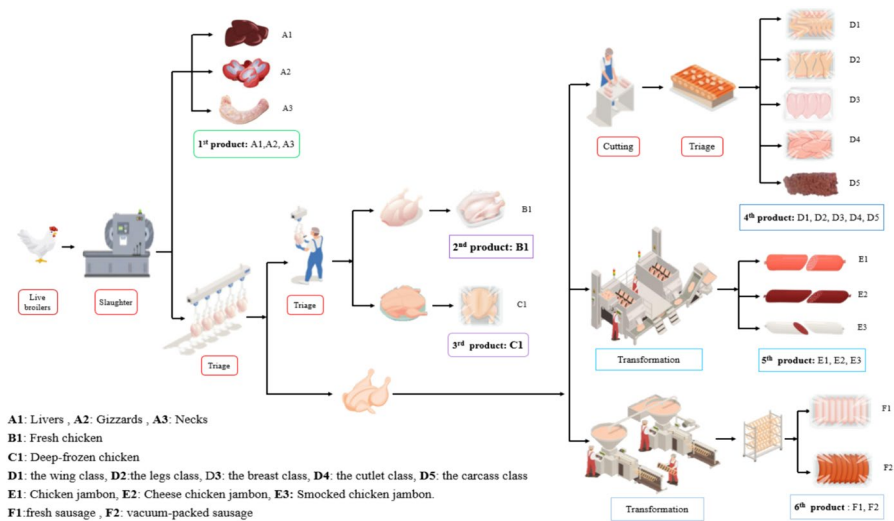


Fig. 2 The chicken processing into by-products

- The quantity of live chickens to be supplied is available throughout the planning horizon.
- The average rate of downgraded chickens is around 10%, according to the statistics of the poultry group. The downgraded chickens are used to produce products 4, 5 and 6.
- The specialized cold room for deep-freezing and storing deep-frozen chicken has a large capacity (considered infinite).

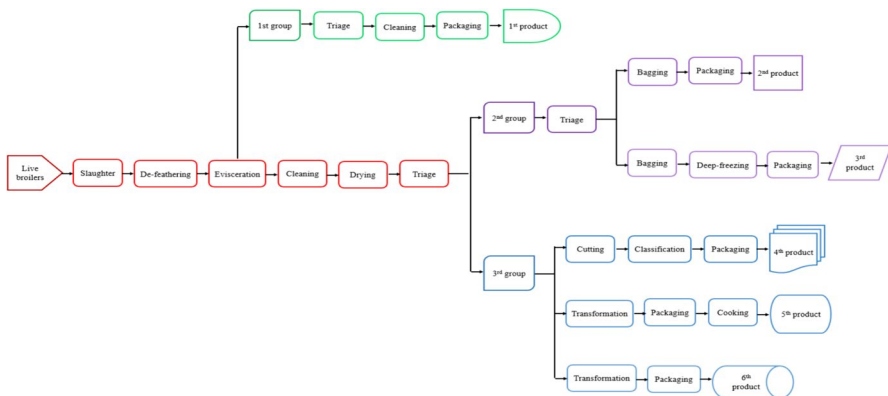


Fig. 3 The different phases of the production process

4.1 Parameters

Indices

- t The set of planning periods $t \in T$
- t' The set of working modes $t' \in T'$
- i The first product set $i \in I$
- j The fourth product set $j \in J$
- k The fifth product set $k \in K$
- l The sixth product set $l \in L$

Data

- Cmx^t Maximum production capacity of the unit for each period t
- Cmn^t Minimum production capacity of the unit for each period t
- $Nou1$ The number of workers in the slaughtering phase.
- $Nou2$ The number of workers in the production phase of the second and third products.
- $Nou3$ The number of workers in the chicken transformation phase for the production of the 4th, 5th, and 6th products.
- $Tt1$ The work rate of the workers in the slaughtering phase.
- $Tt2$ The work rate of the workers in the production phase of the second and third products.
- $Tt3$ The work rate of the workers in the chicken transformation phase for the production of the 4th, 5th, and 6th products.
- Wab_i Weight of slaughter by-products (1st product)
- $Pab_i^{t'}$ Sales price of 1st product (i) for each sub-period t' .
- $Pvpcn$ Sales price of deep-frozen chicken (3rd product).

Fig. 4 The production process of the first group



Fig. 5 The production process of the second group

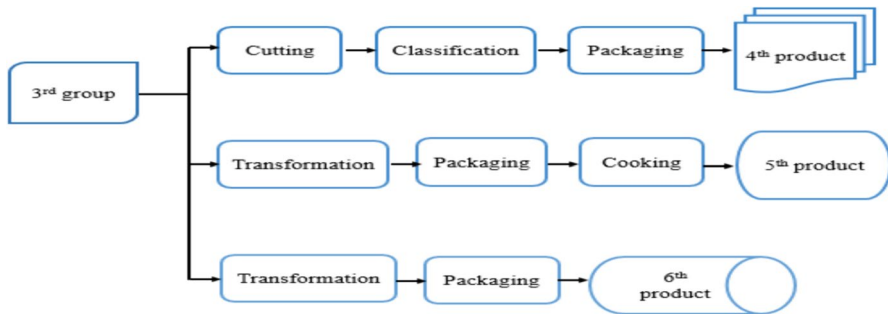
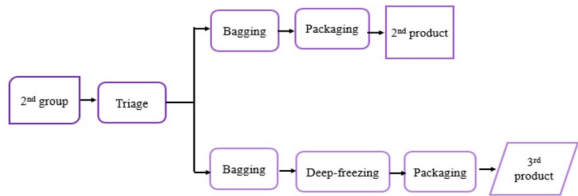


Fig. 6 The production process of products 4, 5, and 6 from the third group

P_{vpfr} Sales price of fresh chicken (2nd product).

P_{vspd1_j} Sales price of 4th product (j).

P_{vspd2_k} Sales price of 5th product (k).

P_{vspd3_l} Sales price of 6th product (l).

Cap_{mxs} Maximum capacity of the storage chamber for the 3rd product.

D_{pcn}^t Demand for 3rd product in each period t.

D_{pfr}^t Demand for 2nd product in each period t.

F_v Meat factor for the production of the sixth product.

F_p Weight factor for the production of the fifth product.

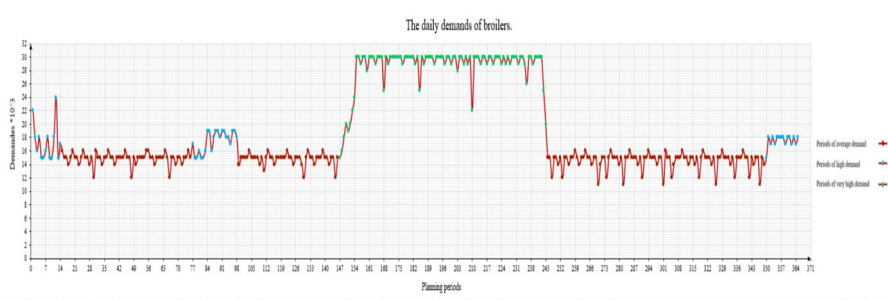


Fig. 7 The daily demands of broilers

- F_a Addition factor for the production of the fifth product.
- F_{pk} Weight factor of a fifth product unit.
- N_{eb_j} Number of items for each piece of the fourth product (j) in a tray.
- N_b Number of trays per box
- F_{os} Bone factor in a chicken.
- F_{dos} Carcass factor of a chicken.
- F_{osp} Bone factor in a chicken breast.
- N_{car} Number of carcasses in a box

The costs

- Cab^t Unit cost of slaughtering and cleaning chickens in period t .
- $Cprab_j^t$ Unit cost of cleaning and packaging the first product i in period t .
- Cpr^t Unit cost of sorting chickens to produce the 2nd and 3rd products in period t .
- $Epcn$ Unit cost of bagging and packaging of 3rd product.

Table 2 The working hours for each work mode

Work mode	Regular	Regular + overtime	Regular + second shift
Working hours	8H à 16H	16h à 20H	16H à 00H

$Epfr$	Unit cost of bagging and packaging of 2nd product.
$Cprd1^t$	Unit cost of production and packaging of the 4th product in period t .
$Cprd2^t$	Unit cost of production and packaging of 5th product in period t .
$Cprd3^t$	Unit cost of production and packaging of 6th product in period t .
$Cspcn$	Unit cost of storing the 3rd product
$Chr1^{tt'}$	Cost of an equivalent worker in man-hours in the slaughter phase for each sub-period t' in period t
$Chr2^{tt'}$	Cost of an equivalent worker in man-hours in the production phase of 2nd and 3rd products for each sub-period t' in period t .
$Chr3^{tt'}$	Cost of an equivalent worker in man-hours in the chicken processing phase for the production of the 4th, 5th, and 6th products for each sub-period t' in period t .
$Cl^{t'}$	Start-up cost of the work teams for each sub-period t'

Decision variables

$X^{tt'}$	$\begin{cases} 1 & \text{start of production at sub - period } t' \text{ in period } t \\ 0 & \text{else} \end{cases}$
$Y^{tt'}$	$\begin{cases} 1 & \text{the choice of the sub - period } t'' \\ 0 & \text{else} \end{cases}$
$Nhr1^t$	Number of working hours per worker in the chicken slaughter phase in period t .
$Nhr2^t$	Number of working hours per worker in the production phase of the second and third products in period t .
$Nhr3^t$	Number of working hours per worker in the processing phase of for the production of the 4th, 5th, and 6th products in period t .
$Qe^{tt'}$	The quantity of live chickens entering the slaughter phase for chicken production for each sub-period t' in period t .
$Qab_i^{tt'}$	The quantity of 1st product (i) for each sub-period t' in period t .

$Pe^{tt'}$	Number of whole chickens produced for each sub-period t' in period t .
$Pd^{tt'}$	Number of chickens for cutting for each sub-period t' in period t .
$Pfr^{tt'}$	Number of fresh chickens (2nd product) for each sub-period t' in period t .
$Pcn^{tt'}$	Number of deep-frozen chickens (3rd product) for each sub-period t' in period t .
$Pcnt^t$	Total number of deep-frozen chickens (3rd product) in period t .
$Pcnts^t$	Total number of deep-frozen chickens stored (3rd product) in period t .
$Pcntv^t$	Number of deep-frozen chickens sold (3rd product) for each in period t .
$Pd1^{tt'}$	Number of chickens for the production of the 4th product for each sub-period t' in period t .
$Pd2^{tt'}$	Number of chickens for the production of the 5th product for each sub-period t' in period t .
$Pd3^{tt'}$	Number of chickens for the production of the 6th product for each sub-period t' in period t .
$Spd1_j^{tt'}$	The quantity of 4th products (j) for each sub-period t' in period t .
$Spd2_k^{tt'}$	The quantity of 5th products (k) for each sub-period t' in period t .
$Spd3_l^{tt'}$	The quantity of 6th products (l) for each sub-period t' in period t .

4.2 The objective function

The objective of this study is to maximise the profit of the slaughtering and processing unit of chickens into different products. The unit has a range of different products to satisfy the market demand: the “1st product” ($Qab_i^{tt'}$), the “2nd product” is fresh chicken ($Pfr^{tt'}$), the “3rd product” is deep-frozen chicken ($Pcntv^t$), as well as another set of different products: The “4th product” is chicken cuts ($Spd1_j^{tt'}$), and the “5th and 6th products” are respectively: chicken jambon ($Spd2_k^{tt'}$), and sausage ($Spd3_l^{tt'}$).

The objective function is used to calculate the unit’s profit by taking into account all costs: the cost of starting up the work teams for each sub-period ($Cl^{tt'}$), the cost of slaughtering (Cab^t), the cost of sorting, cleaning and packaging the “1st product” ($Cprab_i^t$), the cost of cleaning and sorting the chickens (Cpr^t), the cost of packaging the “2nd product” (fresh chickens) (Epr) the cost of packing the “3rd product” (

deep-frozen chickens) (**Epcn**), the cost of storing deep-frozen chickens (**Cspcn**), the cost of producing and packing the “4th product” (**Cprd1^t**), the cost of producing the “5th product” (**Cprd2^t**), the cost of producing the “6th product” (**Cprd3^t**), as well as the unit costs for workers for each sub-period: slaughter workforce (**Chr1^t**), chicken cleaning and sorting workforce (**Chr2^t**), production workforce for the “4th, 5th and 6th products” (**Chr3^t**). The function is formulated and given the following formula (1):

$$\begin{aligned}
 \max F = & \sum_i \sum_t \sum_{t'} Qaba_i^{t \ t'} * Pvaba_i^{t \ t'} + \sum_i \sum_t Pfr^{t \ t'} * Pvpfr + \\
 & \sum_i Pcnt^{t'} * Pvpcn + \sum_j \sum_t \sum_{t'} Spd1_j^{t \ t'} * Pvspd1_j + \\
 & \sum_k \sum_t \sum_{t'} Spd2_k^{t \ t'} * Pvspd2_k + \sum_l \sum_t \sum_{t'} Spd3_l^{t \ t'} * Pvspd3_l + \\
 & (\sum_i \sum_{t'} Cl^{t'} * Y^{t'} + \sum_i \sum_{t'} Qe^{t \ t'} * Cab^l + \sum_i \sum_t \sum_{t'} Qaba_i^{t \ t'} * Cprab_i^{t'} + \\
 & \sum_i \sum_{t'} Pe^{t \ t'} * Cpr^{t'} + \sum_i \sum_t Pfr^{t \ t'} * Epfr + \sum_i Pcnt^{t'} * Epcn + \\
 & \sum_i Pcnts^{t'} * Cspcn + \sum_j \sum_t \sum_{t'} Spd1_j^{t \ t'} * Cprd1^{t'} + \\
 & \sum_k \sum_t \sum_{t'} Spd2_k^{t \ t'} * Cprd2^{t'} + \sum_l \sum_t \sum_{t'} Spd3_l^{t \ t'} * Cprd3^{t'} + \\
 & \sum_i \sum_{t'} Chr1^{t \ t'} * Nhr1^{t \ t'} + \sum_i \sum_{t'} Chr2^{t \ t'} * Nhr2^{t \ t'} + \sum_i \sum_{t'} Chr3^{t \ t'} * Nhr3^{t \ t'})
 \end{aligned} \tag{1}$$

4.3 Under the constraints

$$Y^{t'} = 1 \quad \forall t' = 1 \tag{2}$$

$$\sum_{t' > 1} Y^{t'} \leq 1 \tag{3}$$

$$X^{t \ t'} \leq Y^{t'} \quad \forall t : 1 < t \leq 8 \ \forall t' \tag{4}$$

$$Qe_i^{t \ t'} \leq Cmx^{t \ t'} * X^{t \ t'} \quad \forall t \ \forall t' \tag{5}$$

$$Qe_i^{t \ t'} \geq Cmn^{t \ t'} * X^{t \ t'} \quad \forall t \ \forall t' \tag{6}$$

$$Nou1 * Ti1 * Qe^{t \ t'} = Nhr1^{t \ t'} \quad \forall t \ \forall t' \quad (7)$$

$$Nou2 * Ti2 * Pe^{t \ t'} = Nhr2^{t \ t'} \quad \forall t \ \forall t' \quad (8)$$

$$Nou3 * Ti3 * Pd^{t \ t'} = Nhr3^{t \ t'} \quad \forall t \ \forall t' \quad (9)$$

$$Qab_i^{t \ t'} = Qe^{t \ t'} * Wab_i \quad \forall i \ \forall t \ \forall t' \quad (10)$$

$$Pe^{t \ t'} = 0.9 * Qe^{t \ t'} \quad \forall t \ \forall t' \quad (11)$$

$$Pd^{t \ t'} = 0.1 * Qe^{t \ t'} \quad \forall t \ \forall t' \quad (12)$$

$$Pe^{t \ t'} = Pcn^{t \ t'} + Pfr^{t \ t'} \quad \forall t \ \forall t' \quad (13)$$

$$Pfr^{t \ t'} = 0 \quad \forall t \ \forall t' : t' > 1 \quad (14)$$

$$\sum_{t'} Pfr^{t \ t'} = Dpfr^t \quad \forall t \quad (15)$$

$$Pcnts^t = Ssp \quad \forall t = 1 \quad (16)$$

$$Pcnts^t = Ssf \quad \forall t = 8 \quad (17)$$

$$Pcnts^t \geq Ss \quad \forall t \quad (18)$$

$$Pcnt^t = \sum_{t'} Pcn^{t \ t'} \quad \forall t \quad (19)$$

$$Pcnts^{t-1} + Pcnt^{t-1} - Pcntv^t = Pcnts^t \quad \forall t : 1 < t \leq 8 \quad (20)$$

$$Pcnts^{t-1} \geq Pcntv^t \quad \forall t : 1 < t \leq 8 \quad (21)$$

$$Pcntv^t = Dpcn^t \quad \forall t \quad (22)$$

$$Pcnts^t + Pcnt^t \leq Capmxs \quad \forall t \quad (23)$$

$$Pd1^{t \ t'} = 0.4 * Pd^{t \ t'} \quad \forall t \ \forall t' \quad (24)$$

$$Pd2^{t \ t'} = 0.4 * Pd^{t \ t'} \quad \forall t \ \forall t' \quad (25)$$

$$Pd3^{t \ t'} = 0.2 * Pd^{t \ t'} \quad \forall t \ \forall t' \quad (26)$$

$$Spd1_j^{t \ t'} = Pr * Pd1^{t \ t'} / Neb_j \quad \forall j : j \leq 4 \forall t \ \forall t' \quad (27)$$

$$\begin{aligned} Spd1_j^{t \ t'} &= Pd1^{t \ t'} * Fdos + 0.05 * Pd1^{t \ t'} * Fosp + \\ &Pd2^{t \ t'} * Fos + Pd3^{t \ t'} * Fos \end{aligned} \quad \forall j : j = 5 \forall t \ \forall t' \quad (28)$$

$$Spd2_k^{t \ t'} = Np5 * Pd2^{t \ t'} * Fa * Fp / Fpk \quad \forall k \ \forall t \ \forall t' \quad (29)$$

$$Spd3_l^{t \ t'} = Np6 * Pd3^{t \ t'} * Fv \quad \forall l \ \forall t \ \forall t' \quad (30)$$

$$\begin{aligned} Qe^{t \ t'} &\in R^+ \quad Qab_i^{t \ t'} \in R^+ \quad Pe^{t \ t'} \in R^+ \quad Pd^{t \ t'} \in R^+ \\ Pfr^{t \ t'} &\in R^+ \quad Pcn^{t \ t'} \in R^+ \quad Pcnt' \in R^+ \quad Pcnts^t \in R^+ \\ Pcnv^{t \ t'} &\in R^+ \quad Pd1^{t \ t'} \in R^+ \quad Pd2^{t \ t'} \in R^+ \quad Pd3^{t \ t'} \in R^+ \quad \forall i \ \forall j \ \forall k \\ &\quad \forall l \ \forall t \ \forall t' \\ Spd1_j^{t \ t'} &\in R^+ \quad Spd2_k^{t \ t'} \in R^+ \quad Spd3_l^{t \ t'} \in R^+ \\ X^{t \ t'} &\in \{0, 1\} \quad Y^{t \ t'} \in \{0, 1\} \end{aligned} \quad (31)$$

Constraint (2) ensures the mandatory choice of working the unit in regular time for the whole planning horizon, the constraints (3), (4) impose the choice of one of the work modes: overtime or hiring of the 2nd shift, in case the regular hours are not sufficient to meet the demand. Constraints (5) and (6) ensure that the number of live chickens entering the slaughterhouse must be appropriate to the production capacity of the slaughterhouse during the entire planning horizon and for each work mode (sub-period) t' . and for the Constraints (7), (8) and (9) calculate the number of working hours of the workers in each production phase during the whole planning horizon.

Constraint (10) calculates the quantity of the 1st product during the whole planning horizon and for each work mode (sub-period) t' . Constraints (11), (12) calculate the number of chickens after sorting: whole chickens and chickens for the production of processed products over the entire planning horizon. Constraint (13) represents the conservation of the quantity flows of whole chickens, fresh

chickens (2nd product), and frozen chickens (3rd product) over the entire planning horizon. Constraint (14) imposes the production of the “2nd product” (fresh chickens) only with the regular work mode for each period t during the whole planning horizon. Constraint (15) ensures that the demand for the “2nd product” is fully satisfied throughout the planning horizon t .

The Constraints (16), (17), and (18) represent respectively the initial and final safety stock, as well as the safety stock of the “3rd product” (deep-frozen chickens) for all periods of the planning horizon t . Constraint (19) calculates the total quantity of “3rd product” (deep-frozen chickens) produced for each period of the planning horizon. Constraints (20) and (21) represent the conservation of the flows of quantities of deep-frozen chickens produced, stored, and sold for the entire planning horizon. Constraint (22) ensures that the demand for “3rd product” is fully satisfied for each period of the planning horizon. Constraint (23) ensures that the maximum capacity of the “3rd product” storage room is respected during the entire planning horizon.

Constraints (24), (25), and (26) respectively calculate the number of chickens needed for the production of the following three types of products: the “4th product”, the “5th product”, and the “6th product”. Constraints (27), (28) calculate the quantity produced of the “4th product” for each period t and each work mode (sub-period) t' . Constraints (29), (30) calculate respectively the quantity produced of the “5th product” and the “6th product” for each period t and each work mode (sub-period) t' . Finally, constraint (31) represents the nature of the real and binary decision variables.

5 Numerical application

The numerical parameters used for the proposed modelling are collected from the poultry company as well as the surveys carried out in the field (at the various slaughterhouses) and in the concerned sector. this data is illustrated in Fig. 7 where the estimated customer demands for the entire planning horizon are plotted. As well, Table 2 represents all costs and expenses related to the production and processing of live chickens into multi-products. To solve the studied problem, the proposed model was implemented under the Cplex solver. The aim is to calculate, on one hand, the optimal quantities to be supplied to satisfy the daily demands and, in the other hand, the adequate work schedule for all planning periods.

5.1 The demands of the study horizon

According to Fig. 7, the received demands vary considerably over the year, from one period to another and from one season to another.

To establish a plan model for this system based on the history of the last five years for confidential reasons of the company, demand forecasts for the year 2022 are presented in Fig. 4. For the prediction tool, the Eviews software is used. The demands are grouped into three types. There are periods of average demands,

periods of high demands, and periods of peaks (very high demands). Fluctuations in demands are related to local events such as holidays, festive events and weddings. The demands are subject to the character of uncertainty since they are of random nature that can be interpreted by a uniform distribution with a definite mean and standard deviation.

5.2 The costs

To implement the proposed model in the Cplex solver, all the different costs in the system were considered. The production unit has several costs: costs related to the slaughter operation, costs related to the cleaning operation and costs related to the transformation of chickens into multi-products. The different costs considered are represented in the following table (Table 3) for the different working modes (Regular, Regular + overtime, Regular + second shift).

5.3 Data related to the production unit

Regarding the information related to the production lines, each of them is characterized by minimum and maximum capacities and a certain number of workers according to the type of operation that will be performed. For example, a number of workers are assigned to the slaughtering phase, others to the preparation of 2nd and the 3rd products, and the third group of workers is involved in the preparation of the processed products (4, 5 and 6). Thus, the number of workers varies from one mode to another. The summary of all the information is shown in the following table (Table 4).

6 Results and discussion

As mentioned above, to test the efficiency of the proposed model, it has been implemented under the Cplex solver. The obtained results are represented in the table and figures above (Table 5), (Fig. 8), (Fig. 9). The proposed model is used in the first step to determine the work mode (regular, regular + overtime, or regular + 2nd shift) for all horizons. The second step is to determine which quantity to supply to satisfy the received demands for products 2 and 3. Then, the maximization of the unit's profit. The third part deals with "product 3" (deep-frozen chicken) since deep-frozen chickens have somewhat particular management, as specified in Sect. (3.2).

6.1 The work schedule

The obtained results allow establishing an adequate work schedule for all the studied periods, which is represented in the following table (Table 5):

Table 3 The different considered costs according to the work mode

The costs	The value of each work mode		
	Regular	Regular + overtime	Regular + 2nd shift
Slaughtering costs	30	30	30
Preparation cost of the 1st product	10	10	10
Preparation cost of the 2nd and 3rd products	50	50	50
Preparation cost of the 4th, 5th, and 6th products	20	20	20
Packaging cost of the 2nd product	24	24	24
Packaging cost of the 3rd product	30	30	30
Storage costs of the 3rd product	01	01	01
Cost of an equivalent worker in man-hours in all production phases: slaughter, and preparation of products 1, 2, 3, 4, 5 and 6	0.16	0.2	0.18

From the obtained results, despite the variation in demands, the proposed modelling has effectively solved the problem since it allowed for the creation of a work schedule specifying the work mode for each planning period while respecting the ergonomic aspect of human beings. It is observed that, all three modes were used throughout the horizon to meet the various demands, which allowed the slaughter unit to be organized since it is the key element in the poultry supply chain. In fact, the slaughter unit has the heavy responsibility of managing inputs and outputs as much as possible, avoiding massive overstocking and stock-outs. Moreover, the perishable nature of fresh food (white meat) further complicates the smooth running of operations in these units. In this amalgam of constraints, the proposed model was able to draw up an optimal plan that synchronizes the work teams according to fluctuations in demand while minimizing the various expenses.

The work was carried out for a planning horizon of one year divided into 54 periods where each period is governed by a demand that differs from one period to another. The resulting solution allowed for making assignments between the different periods and work modes. This gave a well-structured work schedule. For the periods (T3 to T11, T15 to T21, and T36 to T50), the company adopted a regular mode since the demands were of average values. While for the periods (T1, T2, T12, T13, T14, T51, T52 and T53), the established model suggested the application of the second mode (regular + overtime). Namely that, for the overtime, the workers already recruited will be called upon to carry out a few hours over and above their allocated time. For the periods (T22 to T35) which correspond to the periods of local festivities and weddings, where the demand is very high, a completely external shift is employed.

The proposed modelling determined when to employ the second shift, knowing that the second shift cannot be employed for a single day or a single horizon. The second shift has to work for a fixed period of a minimum duration. This shift is recruited with contracts where it can be called in each time as needed.

Table 4 The data related to the production unit according to the work mode

The data	The value of each work mode		
	Regular	Regular + over-time	Regular + 2nd shift
Minimum production capacity of the unit	16,000	8,000	16,000
Maximum capacity of the unit	5,000	1,500	5,000
Number of workers in the slaughtering and preparation of 1st product	20	20	20
Number of workers in the slaughtering and preparation of 2nd and 3rd products	12	12	12
Number of workers in the slaughtering and preparation of the 4th, 5th and 6th products	08	08	08
The work rate of the workers in all production phases	0.0004	0.0004	0.0004
Maximum capacity of the cold room	30,000 chickens		

Table 5 The obtained work schedule

Working modes	Regular	Regular + over time	Regular + 2nd shift
The chosen periods	T3, T4, T5, T6, T7, T8, T9, T10, T11, T15, T16, T17, T18, T19, T20, T21, T36, T37, T38, T39, T40, T41, T42, T43, T44, T45, T46, T47, T48, T49, T50	T1, T2, T12, T13, T14, T51, T52, T53	T22, T23, T24, T25, T26, T27, T28, T29, T30, T31, T32, T33, T34, T35

These contracts are different in terms of payment, which influences the selling price of the white meat, but it is still a typical management to satisfy the demand and raise the problem of production instability, stock shortages, and over-stocking of frozen chickens. The obtained results allowed for prediction whether the demand will be met just by working for regular operation, or whether there is a need to work overtime or to use the second shift altogether, and this applies to all studied planning horizons.

6.2 Production quantity planning

The quantities of incoming chickens undergo several processing steps, each quantity is divided into three groups ($\mathbf{Pfr}^{tt}$, $\mathbf{Pcnt}^t$, $\mathbf{Pd}^{tt}$). Through the obtained results represented in the graph of Fig. 8, the optimal quantity that satisfies the forecasted demands ($\mathbf{Pfr}^{tt}$, $\mathbf{Pcnt}^t$) and the quantity destined for cutting ($\mathbf{Pd}^{tt}$) are determined. This is applied over the entire planning horizon.

The results show the effectiveness of the model since it permitted to give the proposed planning, which is accompanied by the determination of the optimal production quantities of each type of product to satisfy daily market demand.

In addition, the production for the period will satisfy the demand for “product 2” (fresh chickens) on the one hand, and on the other, will be destined for deep freezing and storage for subsequent periods. For example, for a demand of fresh chickens (2nd product) around 8,700 chickens, the slaughter unit will produce 15,389 chickens. This quantity will be divided into three quantities: 8,700 chickens to satisfy the fresh chickens demand, 5,150 chickens destined for deep-freezing to obtain the “3rd product” to meet further demands, and 1,539 chickens destined for the transformation to obtain the “4th, 5th, and 6th products”.

6.3 Deep frozen chicken management

This phase, which concerns deep-frozen chicken management, is one of the most complex phases of the system to manage. On the one hand, there is a stock to be maintained, on the other hand, there is an incoming quantity that will undergo deep freezing for a determined period before being sent to the storage areas, as well as an outgoing quantity to satisfy demand.

The particularity of this management is the notion of time since a deep-frozen chicken requires a minimum of 48 h of freezing before being sent. This makes this phase complicated and requires somewhat special management since there are entries, exits and stocks that are in constant rotation daily. This can be interpreted as a very dynamic and highly changeable flow, which explains why it is necessary to have quantities in stock at the beginning and the end of each planning period.

This additional storage constraint at the beginning and the end of each period further complicates the planning. However, its presence is mandatory because the massive stock at the beginning of the period is used to cover the demands for this product for the next two days. This explains why, on the first day of the chosen period, the stock is around 11,000 chickens, which corresponds to the satisfaction of the demands of the first two days (d1 and d2) of the same period with a quantity of safety stock to satisfy unpredictable demands. The sold quantities lead to a decrease in stocks, but in return, on the same days, the cold room will receive new quantities to satisfy further demands after the 48 freezing days.

From the results represented in Fig. 9, it is observed that the proposed modelling allowed for the determination of the incoming quantity, the quantities in stock of deep-frozen chicken, the quantities that are in the transitory phase of deep-freezing, and then the outgoing quantities. In this way, the overall quantity that circulates in the system to satisfy the different variants of the by-products was determined in the first phase. In a second phase, once the decisional determination of the quantities was made, the model permitted to select the work mode to be chosen to maintain a more or less stable production during the whole year.

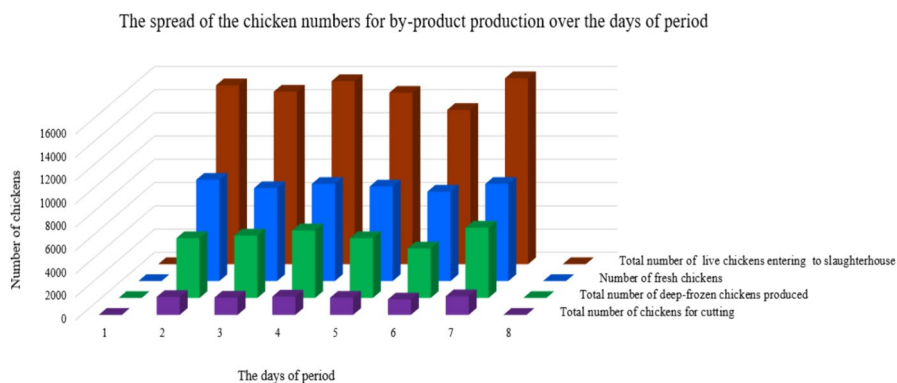


Fig. 8 The spread of the chicken number for by-products production

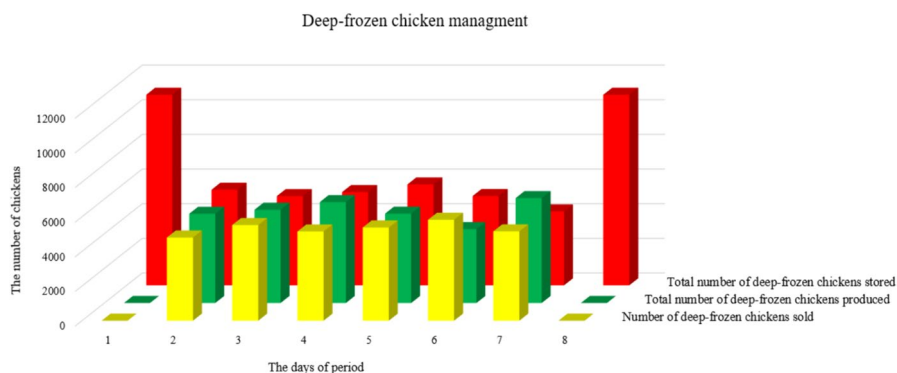


Fig. 9 The deep-frozen chicken management

7 Comparison of the unit management before and after planning

To visualise the economic and decisional impact of the proposed tactical-operational planning approach, adapted to the studied meat processing unit. A managerial comparison between the traditional organisation and the introduction of the proposed new approach and their impact on the different considered points within the studied poultry chain: work schedule, production management, demand satisfaction rate, and stock management of deep-frozen chickens.

7.1 Work schedule

On the organizational level, the work previously operated with a single shift corresponding to a maximum 8 h per day spread over a period of six days per week. This way of working inflicted failures on the company in terms of client satisfaction and meeting demands. In other words, in the majority of cases, the

quantities of demand were not satisfied, whereas the capacity of resources was exploitable and allowed more use, except that the bad management and shutdown of the production units after 8 h generated a situation of cumulative breakdown over the different periods. To address the problem and respond effectively to the deficits of this work regime, improvements on the work schedule were introduced throughout the planning horizon after adding overtime or recruiting a seasonal team to ensure adequate production to the given demands, as shown in Table 3, the satisfaction rate of product and by-product demands reached 100% by adopting the proposed new approaches (more details will be given in the next Sect. 7.3).

7.2 Production management

The production plan has been significantly improved by the implementation of the proposed approach since it allowed determining the optimal quantities to be produced to meet the given demands on the one hand. Moreover, on the other hand, an optimal distribution and allocation between material and human resources while respecting the machinery capacities and the ergonomic aspects of the workers.

7.3 Demand satisfaction rate

The main produced products in the studied meat-processing unit are fresh chicken and deep-frozen chicken. Where their production during the primary operation was made in a totally anarchic way under the principle of pushed flow. In this case, the unit works in mass production, which consists in filling the storage room first and then satisfying the demands of fresh chicken ready for sale and marketing. Except that this psychology of functioning was not convenient since in the majority of cases, the fresh chicken demand was not satisfied, despite the existence of the capacity to produce more. The majority problem was the poor production management, which created an imbalance between the production of deep-frozen and fresh chicken, resulting in the loss of customer orders. Knowing that sales are carried out within this company according to different principles relating to each type of product (fresh chicken, deep-frozen chicken).

7.3.1 The deep-frozen chicken

After filling the cold rooms, the unit starts to serve the demands from the quantity in stock available until the exhaustion of this stock without taking into consideration the instantaneous replenishment which allows to assure the balance between the incoming and outgoing flows of the cold room. This way generates stock shortages because there was no preliminary plan to determine the moment or the point of order from which production had to be restarted. Once the company is confronted with the stock shortage, it is at this moment that the production orders are re-launched for a massive filling of the cold rooms. And this

situation is repeated several times during the working horizons of this company, as presented in the following Fig. 10.

Obviously, this affects sales with the dissatisfaction of the demands related to the stock rupture periods. Moreover, the acquisition of new quantities after the break requires a minimum duration of 48 h, which influences negatively and affects in a considerable way on profit and credibility of the company. By finding itself in several deficit situations as presented in the periods (T4, T5, T6, T12, T18, T21, T22, T23, T24, T27, T28, T33, T34, T37, T38, T39, T40, T50, T51, and T52) as presented in Fig. 11.

7.3.2 The fresh chicken

Regarding the production and sale of fresh chicken, the majority of sales were not satisfied because this product is the subject of a second priority. Knowing that the main priority is given to the frozen product in terms of production, but this situation does not reflect reality. The daily consumption from the history of demand presents the same priority for both products, that is, the perishability and poor management of the company force the company to work more on the frozen product than the fresh product. This explains the decline in the quantity sold and the dissatisfaction of the daily demand for fresh chicken represented in Fig. 12, since a minority quantity was reserved for daily sale (fresh product) and the rest was automatically intended to fill the cold rooms to ensure the best conservation of the production of this product.

This work aims to reorganize the quantity in stock and the quantities of fresh products to meet the given demands. That said, to begin, a forecast plan of the demands had been determined from sales history, which is presented in Fig. 3. Based on this forecast plan, it was possible to realize a reallocation and reorganization of production by adopting the various modes proposed: regular mode; regular mode+overtime; and regular mode+second team to meet the demand for fresh chicken on the one hand and on the other side fill cold chambers instantly with quantities of deep-frozen chicken to keep the balance between the flow of outgoing and incoming quantities and subsequently eliminate rupture and overstocking at the same time.

Through the proposed plan. The researchers were able to establish optimal planning by respecting the different technical, managerial, and ergonomic constraints that improve production by increasing capacity and that converge and correspond with forecast demands. As shown in Fig. 13 for deep-frozen fresh chicken and Fig. 14 for fresh chicken.

From Figs. 13 and 14, it can be observed that the results obtained by the proposed approaches are clearly advantageous and favorable for the produced quantities for both types of fresh and deep-frozen products, and effectively respond to variations in demand. The proposed plan was able to determine and restore production to ensure total satisfaction of the demand for both products, which explains the parallel between the two curves of forecasted demand and the quantities sold with a certain tolerable gap. While avoiding recurrent variations of over-storage and storage rupture.

7.4 Management of deep-frozen chicken stocks

Obviously, the differences in produced quantities between before and after operation are very visible on the stock, whose evolution is shown in Fig. 15. As mentioned in the previous Sect. (7.3), the storage of frozen chicken was anarchic and very fluctuating on the market. With the proposed production plan, it has enabled to mitigate this effect, where it allows to propose an adequate stock management that corresponds with the quantities in stock, the forecasted demands, and the quantities sold while ensuring a safe stock to meet unpredictable demands. This allowed to better manage the stock, better consume energy, and not fall into the situation of ruptures, which was explained by the increase in the satisfaction rate of demand.

8 Synthesis

According to the considered benchmark, this semi-automatic meat processing unit previously opted to work in a fixed way without considering variations in the decisions to be modified. It had only one shift and kept the production capacity relatively constant over a period of six days a week,

The traditional functioning does not worry about the effect of over storage; the management opted for a filling of the cold chambers to their maximum capacity. Where each period is treated separately, resulting in a lack of integration between the different periods of the year. Following this decision, intervals of periods were perceived when the stock was full, as shown in Fig. 10, and subsequently, the production of deep-frozen chicken will be forced to stop. And in other intervals of periods, the quantities in stock were too low, which led to losing market orders.

Another disadvantage was how to set a competitive selling price and subsequently win other markets if, at first, the production and storage capacities were not effectively included in this production system. As a result, the economic balance sheet was not clear, and the decision-makers followed the selling price in the consumer market without improving the performance of their production system.

These different negative points have contributed considerably to this study, and through these reflections and field analyses, the researchers have distinguished the need

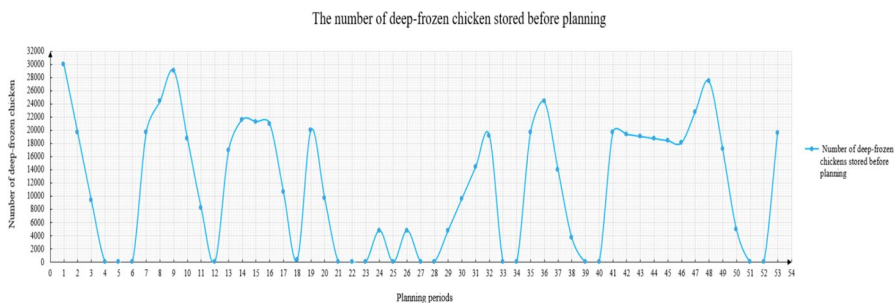


Fig. 10 The number of deep-frozen chicken stored before planning

for improvements, and their research contribution presented aims to offer several solutions and benefits such as.

- The implementation of an integrated approach to tactical-operational planning to better adapt the working mode that increases production capacity and subsequently maximizes profit
- The possibility of offering a better synchronization between the upstream and downstream parts of the chain that optimizes the incoming and outgoing flows by rebalancing the quantities in stock and eliminating over-stocking and stock shortages
- Provide a framework for planning at the level of other semi-automatic meat processing units in a more rational way in terms of energy consumption according to the storage conditions and limits.
- Perfect the cost accounting to improve the research and development plan of this unit.

9 Managerial perspective

The reorganization of production within a chicken slaughterhouse unit has become paramount within poultry chains, considering the increasing consumption of this protein source. Furthermore, the perishable nature of chicken necessitates particular attention to its management, both technically and managerially.

The technical plan involves overseeing process optimization, human resource management, and man–machine allocation, as detailed in the provided study, in line with the objectives set at the tactical-operational level. The second aspect concerns managerial perspectives established by the strategic plan, aiming to define the organization’s long-term goals and the means required to achieve these effectively and efficiently. Particularly for poultry chains, managerial perspective must encompass all aspects of production, marketing, and industry-specific risk management. This entails effective management of technical and human resources, continuous improvement of quality and food safety, and flexibility to

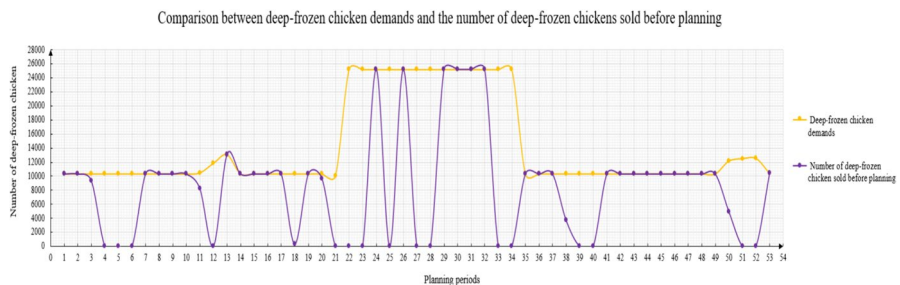


Fig. 11 Comparison between deep-frozen chicken demands and the number of deep-frozen chickens sold before planning

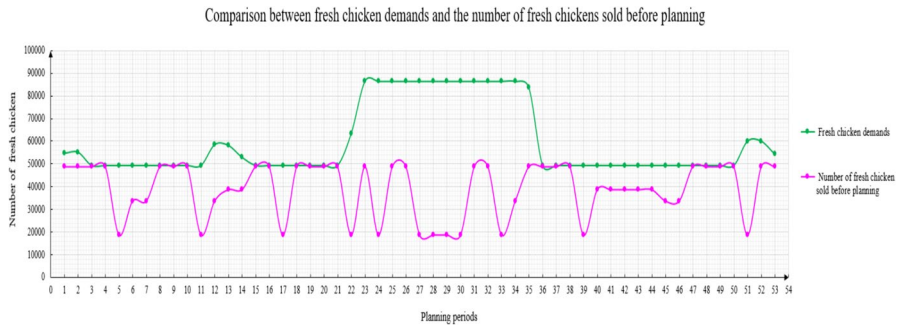


Fig. 12 Comparison between fresh chicken demands and the number of fresh chicken sold before planning

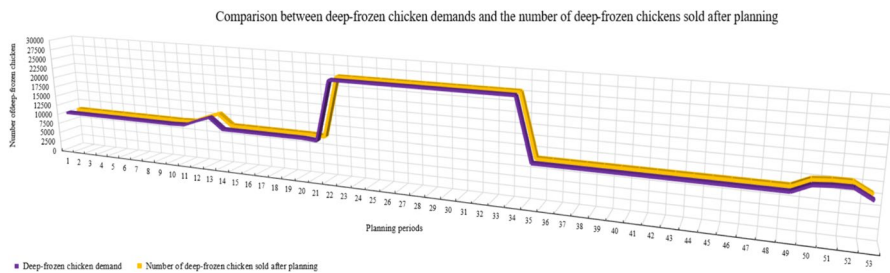


Fig. 13 Comparison between deep-frozen chicken demands and the number of deep-frozen chickens sold after planning

adapt to market changes while adhering to environmental constraints and regulations. These perspectives are detailed in the following points:

- *Logistics Management* Analyzing the company's logistics chain to identify possible improvements in terms of suppliers, logistics, or inventory manage-

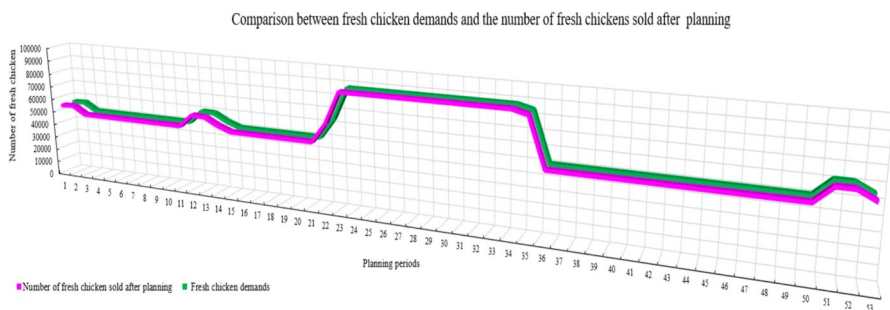


Fig. 14 Comparison between fresh chicken demands and the number of fresh chickens sold after planning

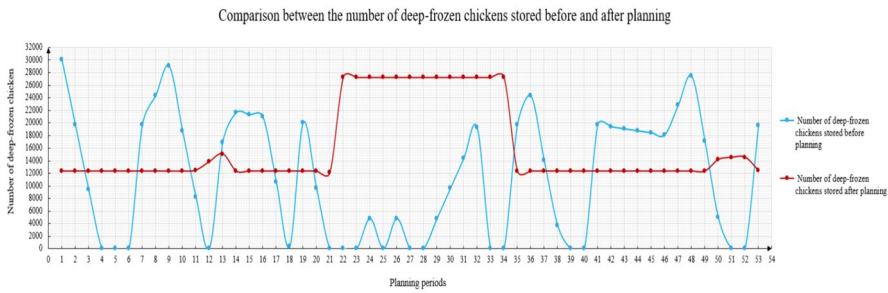


Fig. 15 Comparison between the number of deep-frozen chickens stored before and after planning

ment. Reorganization may aim to make the logistics chain smoother and more profitable.

- **Process Optimization** Examining the production processes of the slaughter unit: slaughtering, cutting, charcuterie production, packaging, and distribution, to identify inefficiencies and bottlenecks. Subsequently, implementing improvements to enhance operational efficiency, reduce costs, and increase productivity.
- **Human Resources Management** Considering the implications of reorganization on personnel, including necessary training for effective unit management, reallocation of workforce to work teams, daily workforce allocation across different modes of work, and ensuring ergonomic aspects for employees. It is crucial to address employees' concerns and maintain high engagement to ensure the transition's success.
- **Quality and Safety Management** Ensuring that food quality and safety standards are upheld throughout the production process. This may involve implementing stricter quality controls and additional training for staff.
- **Business Strategy** Evaluating how production reorganization may affect the company's competitive position in the market. This could include considerations such as expanding production capacity, introducing new products or services, or exploring new markets.
- **Financial Management** Assessing the financial implications of work reorganization at the slaughterhouse unit, including initial costs, potential long-term savings, and profitability forecasts. This may involve a detailed analysis of necessary investments, projected cash flows, and financial performance indicators.
- **Risk Management** Managers must be aware of potential risks for the company, such as fluctuations in live chicken prices, avian diseases, machinery breakdowns, labor shortages, etc. Proactive risk management is necessary to mitigate their impact on poultry chain operations.

By integrating these different managerial perspectives, managers can develop a comprehensive reorganization strategy that maximizes value for the company while

considering the needs and concerns of all stakeholders involved. It should be noted that the adaptation of these managerial perspectives is crucial for all companies in the agri-food industry, whatever the product concerned. Furthermore, they can also be extended to other industries producing red meat, fish, etc. By applying these perspectives, companies can benefit from an integrated and coherent approach to improving their operations and competitiveness, while meeting consumer demands and ensuring the sustainability of their activities.

10 Conclusion

This paper studies a tactical operational production planning problem within a meat processing unit that transforms broiler chickens into multiple products over a multi-period planning horizon. The objective was to develop a work schedule that maximizes the efficiency of this unit. This research has introduced a decision-making approach to reorganization and task allocation that determines the optimal quantities to meet given demands for all products.

The proposed work schedule addresses three types of work modes: regular, regular plus overtime, or regular plus a second shift. This ensures optimal assignments between work modes and demand satisfaction while providing ease and respect for the ergonomic aspects of workers. The results of the experiments indicate that the total income, the produced quantities flow, as well as the storage quantities, and the breakages encountered during the periods, have improved considerably compared to the classical operation with only one shift, whether for fresh or deep-frozen products.

The developed mathematical model is a generic model that can be adapted to other types of meat processing units intended for the slaughter of animal products from poultry, cattle, and sheep, such as cattle, ducks, sheep, etc., and their transformations into multi-products. The organization of the hourly volume can apply to a wide range of applications and industrial systems, whether agro-food or otherwise, and to all types of workshops where the material and human resources have sufficient capacity for overtime or an additive team. Variations and changes may occur depending on the type and nature of the product to be produced.

The proposed modeling provides better management that helps unit decision-makers make the right decisions regarding the work organization within the unit by adding overtime or recruiting a second external team over a specified period. This approach guaranteed the demand satisfaction of the different products and the maximization of the unit's profit.

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Data availability The data are collected from surveys carried out in the field at the various chicken slaughterhouses.

Declarations

Conflict of interest The authors declare that they have no competing interest.

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

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